

▲ Vercel Deployment Guide

Complete step-by-step guide for deploying AuthiChain to Vercel.

Why Vercel?

Pros:

-  **Easiest deployment** - Git push to deploy
-  **Optimized for Next.js** - Built by Next.js creators
-  **Auto-scaling** - Handles traffic spikes automatically
-  **Global CDN** - Fast loading worldwide
-  **Zero-downtime deployments** - Atomic deployments
-  **Preview deployments** - Test before production
-  **Free SSL certificates** - HTTPS out of the box

Cons:

-  Serverless functions have cold starts
-  10-second function timeout on free tier

Pricing:

- **Hobby:** FREE (perfect for testing & small projects)
 - **Pro:** \$20/month (for production apps)
 - **Enterprise:** Custom pricing
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Prerequisites

Before starting:

-  GitHub account with AuthiChain repo
 -  All API keys configured (see [API_KEYS_SETUP.md](#) (./API_KEYS_SETUP.md))
 -  All 15 Stripe Price IDs created (see [STRIPE_PRICE_SETUP.md](#) (./STRIPE_PRICE_SETUP.md))
 -  Production database ready (Railway, Supabase, or AWS RDS)
 -  Domain name (optional but recommended)
-

Deployment Steps

Step 1: Push Code to GitHub

1. Initialize Git (if not already done):

```
bash
cd /path/to/authichain_premium/app
git init
git add .
git commit -m "Initial commit - AuthiChain production ready"
```

2. Create GitHub Repository:

- Go to github.com/new (<https://github.com/new>)
- Repository name: `authichain`
- Visibility: Private (recommended)
- Don't initialize with README (you already have one)
- Click "Create repository"

3. Push to GitHub:

```
bash
git remote add origin https://github.com/YOUR_USERNAME/authichain.git
git branch -M main
git push -u origin main
```

Step 2: Connect Vercel to GitHub

1. Sign up for Vercel:

- Visit vercel.com (<https://vercel.com>)
- Click "Sign Up"
- Choose "Continue with GitHub"
- Authorize Vercel to access your GitHub account

2. Import Project:

- Click "Add New..." → "Project"
 - Find and select your `authichain` repository
 - Click "Import"
-

Step 3: Configure Project Settings

1. Framework Preset:

- Should auto-detect "Next.js"
- If not, select "Next.js" from dropdown

2. Root Directory:

- If your Next.js app is in `app` folder, set root directory to `app`
- Click "Edit" next to Root Directory
- Enter: `app`

3. Build Settings:

- **Build Command:** `npm run build` (default, keep as is)
- **Output Directory:** `.next` (default, keep as is)
- **Install Command:** `npm install` (default, keep as is)

4. Node.js Version:

- Recommended: `18.x` or `20.x`
- Set in `package.json`:

```
json
{
  "engines": {
    "node": ">=18.0.0"
```

```
}
}
```

Step 4: Configure Environment Variables

This is the most critical step!

1. Open Environment Variables Section:

- In Vercel project settings, expand “Environment Variables”

2. Add ALL Environment Variables:

Click “Add” for each variable:

Database

Name: DATABASE_URL

Value: postgresql://user:password@host:5432/database

Environments: Production, Preview, Development

NextAuth

Name: NEXTAUTH_SECRET

Value: [your-generated-secret]

Environments: Production, Preview, Development

Name: NEXTAUTH_URL

Value: https://your-domain.vercel.app

Environments: Production

Note: Vercel auto-sets NEXTAUTH_URL for preview deployments

Stripe Live Keys

Name: STRIPE_SECRET_KEY

Value: sk_live_51...

Environments: Production

Name: NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY

Value: pk_live_51...

Environments: Production

Name: STRIPE_WEBHOOK_SECRET

Value: whsec_...

Environments: Production

Stripe Test Keys (for Preview/Development)

Name: STRIPE_SECRET_KEY

Value: sk_test_51...

Environments: Preview, Development

Name: NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY

Value: pk_test_51...

Environments: Preview, Development

NFT.Storage

Name: NFT_STORAGE_API_KEY

```
Value: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

```
Environments: Production, Preview, Development
```

App Configuration

```
Name: NEXT_PUBLIC_SITE_URL
```

```
Value: https://your-domain.com
```

```
Environments: Production
```

```
Name: NODE_ENV
```

```
Value: production
```

```
Environments: Production
```

Stripe Price IDs (All 15)

For production environment, add all 15 Price IDs:

```
```
```

```
STRIPE_STARTER_MONTHLY_PRICE_ID=price_...
```

```
STRIPE_STARTER_ANNUAL_PRICE_ID=price_...
```

```
STRIPE_STARTER_LIFETIME_PRICE_ID=price_...
```

```
STRIPE_PROFESSIONAL_MONTHLY_PRICE_ID=price_...
```

```
STRIPE_PROFESSIONAL_ANNUAL_PRICE_ID=price_...
```

```
STRIPE_PROFESSIONAL_LIFETIME_PRICE_ID=price_...
```

```
STRIPE_BUSINESS_MONTHLY_PRICE_ID=price_...
```

```
STRIPE_BUSINESS_ANNUAL_PRICE_ID=price_...
```

```
STRIPE_BUSINESS_LIFETIME_PRICE_ID=price_...
```

```
STRIPE_ENTERPRISE_MONTHLY_PRICE_ID=price_...
```

```
STRIPE_ENTERPRISE_ANNUAL_PRICE_ID=price_...
```

```
STRIPE_ENTERPRISE_LIFETIME_PRICE_ID=price_...
```

```
STRIPE_WHITE_LABEL_MONTHLY_PRICE_ID=price_...
```

```
STRIPE_WHITE_LABEL_ANNUAL_PRICE_ID=price_...
```

```
STRIPE_WHITE_LABEL_LIFETIME_PRICE_ID=price_...
```

```
```
```

1. **Pro Tip:** Use Vercel CLI to bulk add env vars:

```
```bash
```

```
Install Vercel CLI
```

```
npm i -g vercel
```

```
Login
```

```
vercel login
```

```
Link project
```

```
vercel link
```

```
Add env vars from .env.production
```

```
vercel env pull .env.local
```

```
```
```

Step 5: Deploy!

1. Click “Deploy”

- Vercel will:
 - Install dependencies
 - Build your Next.js app
 - Deploy to global CDN
 - Generate preview URL

2. Wait for Build (~2-5 minutes)

- Watch build logs in real-time
- Check for any errors

3. Build Complete!

- You’ll get a URL like: `https://authichain-xxx.vercel.app`
- Click to visit your deployed app

Step 6: Run Database Migrations

 **CRITICAL:** Run migrations after first deployment!

1. Using Vercel CLI:

```
``bash
# In your local project
cd app

# Set production DATABASE_URL locally
export DATABASE_URL="your-production-database-url"

# Run migration
npx prisma db push

# (Optional) Seed database
npx prisma db seed
``
```

1. Verify Migration:

```
bash
npx prisma studio
```

Step 7: Configure Custom Domain

1. Add Domain in Vercel:

- Go to Project Settings → Domains
- Click “Add”
- Enter your domain: `authichain.com`

2. DNS Configuration:

- #### If using Vercel DNS (Easiest):
- Vercel will show you nameservers

- Update at your domain registrar
- Wait for propagation (~24 hours)

If using your own DNS:

- Add A record:

Type: A

Name: @

Value: 76.76.21.21

- Add CNAME for www:

Type: CNAME

Name: www

Value: cname.vercel-dns.com

1. **SSL Certificate:**

- Vercel automatically provisions Let's Encrypt SSL
- Usually takes 1-5 minutes
- Your site will be HTTPS automatically!

2. **Update Environment Variables:**

- Update `NEXTAUTH_URL` to your custom domain
- Update `NEXT_PUBLIC_SITE_URL` to your custom domain
- Redeploy (automatic on env var change)

Step 8: Configure Stripe Webhook

Now that you have a live URL, update your Stripe webhook:

1. **Go to Stripe Dashboard:**

- Navigate to Developers → Webhooks
- Find your webhook endpoint

2. **Update Endpoint URL:**

- Change from temporary URL to:
`https://your-domain.com/api/stripe/webhook`

3. **Test Webhook:**

- In Stripe Dashboard, click "Send test webhook"
- Select `checkout.session.completed`
- Verify it succeeds (200 OK)

Advanced Configuration

Enable Edge Functions (Optional)

For faster response times globally:

1. Create `middleware.ts` in your app root:

```
``typescript
import { NextResponse } from 'next/server';
import type { NextRequest } from 'next/server';
```

```
export const config = {
  matcher: [
    '/api/:path',
    '/nfts/:path'
  ],
};

export function middleware(request: NextRequest) {
  // Add custom headers
  const response = NextResponse.next();
  response.headers.set('x-custom-header', 'AuthChain');
  return response;
}
...

```

1. Deploy - functions will run on Edge Network

Configure Vercel Analytics (Free!)

1. In Vercel Dashboard:

- Go to Project → Analytics
- Click "Enable"

2. Add to `_app.tsx` :

```
````tsx
import { Analytics } from '@vercel/analytics/react';

function MyApp({ Component, pageProps }) {
 return (
 <>

 </>
);
}
...

```

---

## Set Up Logging (Recommended)

### 1. Install Vercel Log Drain:

```
bash
vercel integrations add datadog
or
vercel integrations add logtail

```

### 2. Configure in Dashboard:

- Vercel Dashboard → Integrations
- Choose your logging provider
- Follow setup instructions

---

## Configure Image Optimization

Vercel automatically optimizes images, but you can configure:

### 1. In `next.config.js` :

```
javascript
module.exports = {
 images: {
 domains: [
 'ipfs.io',
 'nft.storage',
 'cloudflare-ipfs.com'
],
 formats: ['image/avif', 'image/webp'],
 },
};
```



## Continuous Deployment

Vercel automatically deploys on every push:

### Production Branch

- **Branch:** `main` (or `master`)
- **Deployments:** Automatic production deployments
- **Domain:** Your custom domain

### Preview Branches

- **Branch:** Any other branch (e.g., `dev`, `staging`)
- **Deployments:** Automatic preview deployments
- **Domain:** Unique Vercel URL per branch

### Pull Requests

- **Every PR** gets a unique deployment
- **Comment** with preview link appears in PR
- **Test** changes before merging



## Testing Production Deployment

### 1. Basic Health Check

```
curl https://your-domain.com/api/health
```

### 2. Test Authentication

1. Visit your site
2. Try registering a new user



3. Verify email confirmation (if enabled)
4. Log in with credentials

### 3. Test Stripe Integration

1. Navigate to pricing page
2. Click “Subscribe” on a tier
3. Use test card: 4242 4242 4242 4242
4. Verify checkout completes
5. Check Stripe Dashboard for payment

### 4. Test NFT Minting

1. Log in as a subscribed user
2. Navigate to “Create NFT”
3. Upload an image
4. Fill in details
5. Mint NFT
6. Verify it appears in your collection
7. Check IPFS URL works

### 5. Performance Test

```
Install Apache Bench
brew install httpd

Load test
ab -n 1000 -c 10 https://your-domain.com/
```



## Monitoring & Logs

### View Real-Time Logs

1. **Vercel Dashboard:**
  - Go to your project
  - Click “Deployments”
  - Select a deployment
  - Click “View Function Logs”

2. **Vercel CLI:**

```
bash
vercel logs
```

### Monitor Performance

1. **Vercel Analytics:**
  - Dashboard → Analytics
  - View:
    - Page views
    - Unique visitors
    - Top pages

- Real User Monitoring scores

## 2. Web Vitals:

- Automatically tracked
- Core Web Vitals (LCP, FID, CLS)

# Troubleshooting

## Build Fails

**Error:** `Module not found` or `Cannot find module`

### Solution:

1. Check `package.json` includes all dependencies
2. Delete `node_modules` and reinstall:

```
bash
rm -rf node_modules package-lock.json
npm install
git commit -am "Update dependencies"
git push
```

## Environment Variables Not Working

**Error:** `process.env.VARIABLE_NAME is undefined`

### Solution:

1. Verify env var is set in Vercel Dashboard
2. Ensure correct environment (Production/Preview/Development)
3. Redeploy:

```
bash
vercel --prod
```

4. For client-side vars, prefix with `NEXT_PUBLIC_`

## Database Connection Timeout

**Error:** `Connection timeout` or `ETIMEDOUT`

### Solution:

1. Verify `DATABASE_URL` is correct
2. Check database is running
3. Add Vercel IPs to database whitelist (if using IP filtering)
4. Use connection pooling:

```
````typescript
// prisma/client.ts
import { PrismaClient } from '@prisma/client';

const globalForPrisma = global as unknown as { prisma: PrismaClient };
```

```
export const prisma =
globalForPrisma.prisma ||
new PrismaClient({
log: ['query'],
});

if (process.env.NODE_ENV !== 'production') globalForPrisma.prisma = prisma;
...
```

Stripe Webhook Not Receiving Events

Error: Webhook shows failed in Stripe Dashboard

Solution:

1. Verify webhook URL is correct (no trailing slash)
2. Check endpoint is publicly accessible
3. Verify webhook signing secret matches
4. Test locally with Stripe CLI:

```
bash
```

```
stripe listen --forward-to https://your-domain.com/api/stripe/webhook
```

5. Check Vercel function logs for errors
-

504 Gateway Timeout

Error: Function execution timeout

Solution:

1. Optimize long-running functions
 2. Upgrade to Pro plan (60-second timeout)
 3. Move heavy processing to background jobs
 4. Use streaming responses for large data
-



Rollback Deployment

If something goes wrong:

1. In Vercel Dashboard:

- Go to Deployments
- Find last working deployment
- Click “...” menu
- Click “Promote to Production”

2. Using CLI:

```
bash
```

```
vercel rollback
```

Performance Optimization

1. Enable Caching

```
// app/api/nfts/route.ts
export const revalidate = 60; // Cache for 60 seconds

export async function GET() {
  // Your API logic
}
```

2. Use Static Generation

```
// app/nfts/page.tsx
export const revalidate = 3600; // Regenerate every hour

export default async function NFTsPage() {
  const nfts = await prisma.nft.findMany();
  return <NFTGrid nfts={nfts} />;
}
```

3. Optimize Images

```
import Image from 'next/image';

<Image
  src={nft.imageUrl}
  alt={nft.name}
  width={500}
  height={500}
  priority={isFeatured}
/>
```

4. Enable Compression

```
// next.config.js
module.exports = {
  compress: true,
  poweredByHeader: false,
};
```

Security Hardening

1. Content Security Policy

```
// next.config.js
const securityHeaders = [
  {
    key: 'Content-Security-Policy',
    value: "default-src 'self'; script-src 'self' 'unsafe-eval' 'unsafe-inline'; style-src 'self' 'unsafe-inline';",
  },
  {
    key: 'X-Frame-Options',
    value: 'DENY'
  },
  {
    key: 'X-Content-Type-Options',
    value: 'nosniff'
  }
];

module.exports = {
  async headers() {
    return [
      {
        source: '/*:path*',
        headers: securityHeaders,
      },
    ];
  },
};
```

2. Rate Limiting

```
// app/api/auth/register/route.ts
import { Ratelimit } from '@upstash/ratelimit';
import { Redis } from '@upstash/redis';

const ratelimit = new Ratelimit({
  redis: Redis.fromEnv(),
  limiter: Ratelimit.slidingWindow(5, '1 h'),
});

export async function POST(request: Request) {
  const ip = request.headers.get('x-forwarded-for');
  const { success } = await ratelimit.limit(ip);

  if (!success) {
    return new Response('Too Many Requests', { status: 429 });
  }

  // Handle registration
}
```

Additional Resources

- [Vercel Documentation](https://vercel.com/docs) (https://vercel.com/docs)
 - [Next.js Deployment Docs](https://nextjs.org/docs/deployment) (https://nextjs.org/docs/deployment)
 - [Vercel CLI Reference](https://vercel.com/docs/cli) (https://vercel.com/docs/cli)
 - [Environment Variables Guide](https://vercel.com/docs/environment-variables) (https://vercel.com/docs/environment-variables)
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Deployment Checklist

Before going live:

- ☐ All environment variables configured
 - ☐ Database migrations run successfully
 - ☐ Custom domain configured and SSL active
 - ☐ Stripe webhook configured and tested
 - ☐ Test user registration and login
 - ☐ Test subscription checkout
 - ☐ Test NFT minting and uploading
 - ☐ Performance monitoring enabled
 - ☐ Error tracking configured
 - ☐ Backup strategy in place
 - ☐ Complete [PRODUCTION_CHECKLIST.md](#) (./PRODUCTION_CHECKLIST.md)
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Congratulations! Your AuthiChain platform is now live on Vercel! 🎉